

Rajdeep Gill

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EDUCATION

Bachelor of Science in Computer Engineering (Co-op)

University of Manitoba

May 2026

Winnipeg, MB

GPA: 4.45/4.5

Relevant Coursework: Data Structures and Algorithms, Systems Engineering Principles, Parallel Programming

EXPERIENCE

Machine Learning Research Student

May 2024 - Dec 2024

Biosensors Research Lab - University of Manitoba

Winnipeg, MB

- Designed and trained a convolutional neural network to predict biological cell viability from 10,000+ diffraction pattern images, achieving **85%** accuracy and demonstrating potential for non-invasive cell viability assessment
- Enhanced particle diffraction centroid detection using computer vision, improving success rate by **17%** (from 75% to 92%) and providing more accurate data for the CNN model
- Developed a Python-based data visualization tool with an interactive GUI to process, filter, and display diffraction patterns and model predictions, enabling researchers to analyze and interpret data efficiently
- Designed and implemented a scalable SQL database to centralize **100,000+** optical diffraction records, replacing a manual file-sharing system, and improving data accessibility and integrity

High-Performance Computing & Data Analysis Student

May 2023 - Aug 2023

Biosensors Research Lab - University of Manitoba

Winnipeg, MB

- Applied dimensionality reduction techniques (t-SNE, PCA) to analyze and cluster datasets, uncovering key correlations for future research directions
- Optimized data analysis workflows by implementing parallel processing, reducing processing time by 90%
- Developed an oscilloscope interfacing program to automate data collection, improving speed and accuracy
- Delivered weekly progress updates, communicating findings and incorporating feedback to improve project outcomes

PROJECTS

Google Docs Clone | React, TypeScript, HTML/CSS, Next.js

- Developed a Google Docs clone with real-time collaboration, user authentication, and document sharing
- Utilized Convex to store and sync document data, allowing multiple users to edit in real-time
- Implemented organization-wide document sharing and user permissions to control access to documents

Financial Dashboard | React, TypeScript, HTML/CSS, Next.js, Hono.js

- Created an expense and income tracking dashboard with user authentication and real-time data visualization
- Implemented a RESTful API to handle user data and transactions, using a PostgreSQL database to store user information

Deaf Curling Accessibility Aid | C, ESP-32, SolidWorks

- Developed a device to assist deaf curlers in receiving instructions from their skip during games
- Designed a PCB and 3D-printed enclosure ensuring durability and portability

PUBLICATIONS

R. Gill, et al., "Machine Learning Classification of Microplastics by Integrating Optical and Dielectrophoresis Features," in *IEEE AP-S/URSI 2025*.

R. Gill, et al., "Dielectric Analysis of Single Biological Cells at Both Microwave and Optical Frequencies," in *IEEE AP-S/URSI 2025*.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, R, SQL, MATLAB

Frameworks & Libraries: React, Next.js, TensorFlow, PyTorch, OpenCV,

Developer Tools: Git, VSCode, Visual Studio, Android Studio, IntelliJ